

## Project Design Phase-II

### Data Flow Diagram & User Stories

|               |                        |
|---------------|------------------------|
| Date          | 6th july 2026          |
| Team ID       |                        |
| Project Name  | Heart Disease Analysis |
| Maximum Marks | 4 Marks                |

**Data Source:** The process begins by collecting heart disease data from reliable sources such as the UCI/Kaggle Heart Disease dataset, hospital records, and patient medical reports. The dataset contains attributes such as age, gender, blood pressure, cholesterol, ECG results, maximum heart rate, chest pain type, and diagnosis.

**Data Preparation:** The collected data is cleaned by removing duplicate records, handling missing values, correcting inconsistencies, and transforming data into a suitable format for analysis. This ensures high-quality and reliable data.

**Data Analysis:** The prepared dataset is analyzed to identify patterns, trends, and relationships among different medical attributes. Statistical summaries help determine the major risk factors contributing to heart disease.

**Visualization Using Tableau:** The cleaned data is imported into Tableau to create interactive dashboards and visualizations such as bar charts, pie charts, line charts, scatter plots, histograms, and KPI indicators. Filters and parameters are added to allow users to explore the data dynamically.

**Insights and Decision Making:** The visualizations help identify high-risk patients, understand disease trends, and evaluate the impact of various health parameters. These insights support healthcare professionals in making faster, accurate, and data-driven clinical decisions.

**Final Output:** The project produces an interactive Tableau dashboard that enables effective monitoring of heart disease risk factors, improves patient care, supports early diagnosis, and enhances overall healthcare decision-making.

# FLOW CHART – HEART DISEASE ANALYSIS USING TABLEAU

Project : Heart Disease Analysis using Tableau

Team ID : HDA-01

Date : 31 January 2025

Team Name : Health Insight Team

## 1. DATA SOURCE



Heart Disease Dataset  
(Kaggle / UCI)



Patient Records,  
Clinical Attributes &  
Test Results



Import Dataset  
(CSV / Excel)



Raw Data  
Loaded in  
Tableau

## 2. DATA PREPARATION



Data Cleaning  
(Handle Missing  
Values)



Remove Duplicates  
& Inconsistent  
Data



Data Transformation  
(Formatting, Derived  
Fields)



Prepared Data  
Ready for  
Analysis

## 3. DATA ANALYSIS



Explore Data  
(Distributions &  
Summary)



Identify Patterns  
& Relationships



Analyze Risk Factors  
(Age, BP, Cholesterol,  
Max Heart Rate, etc.)



Find Key Insights  
& Trends

## 4. VISUALIZATION USING TABLEAU



Connect Data  
to Tableau



Create Visualizations  
(Bar, Pie, Line, Scatter,  
Histogram, etc.)



Add Filters &  
Parameters for  
Interactivity



Design Interactive  
Dashboards

## 5. INSIGHTS & DECISION MAKING



Interpret Results  
& Draw Insights



Support Clinical  
Decision Making



Identify High-Risk  
Patients Early



Improve Treatment  
Planning & Patient  
Outcomes



## FINAL OUTPUT

Interactive Tableau Dashboard  
Providing valuable insights for better  
decision making and improved  
heart disease management.